

# Cait Harrigan, PhD

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I am a PhD candidate at the University of Toronto supervised by [Quaid Morris](#) and [Kieran Campbell](#). I am a Doctoral Fellow at the [UofT Data Sciences Institute](#) and a graduate researcher at the [Vector Institute](#) and [Ontario Institute for Cancer Research](#). I'm passionate about open science, and promoting great mentorship in the sciences.

## RESEARCH INTERESTS

My research focuses on developing machine learning methods to understand cancer evolution through patterns in somatic mutations. I combine computational biology, statistical modeling, and large-scale genomic data analysis to identify the evolutionary constraints that underlie mutation, and identify therapeutically targetable DNA repair deficiencies. With expertise in: cancer genomics, spatial proteomics, deep learning, probabilistic modelling, and programming in python & R in HPC environments.

## EDUCATION

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|----------------------------------------------------------------------------------------------------------------|-----------------|
| <b>PhD in Computer Science, University of Toronto</b><br><i>Supervised by Quaid Morris and Kieran Campbell</i> | 01/21 - present |
| <b>MSc. in Computer Science, University of Toronto</b><br><i>Supervised by Quaid Morris</i>                    | 09/19 - 03/21   |
| <b>Honours BSc. in Computational Biology, University of Toronto</b><br><i>Awarded with distinction</i>         | 09/15 - 06/19   |

## RESEARCH EXPERIENCE

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|-------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Visiting graduate researcher</b><br>The Francis Crick Institute, London, England<br><i>Hosted by Nicholas McGranahan</i>   | 03/24 - 05/24 |
| <b>Visiting graduate researcher</b><br>Memorial Sloan Kettering Cancer Center, New York, USA<br><i>Hosted by Quaid Morris</i> | 05/21 - 09/21 |

## PUBLICATIONS

\* Indicates joint first authorship, † co-corresponding authorship

- 1. Damage and misrepair signatures: compact representations of pan-cancer mutational processes,** [Caitlin F. Harrigan](#), Kieran R. Campbell, Quaid Morris<sup>†</sup>, Tyler Funnell<sup>†</sup>, *bioRxiv*
- 2. Automated registration of spatial expression data scales multimodal integration to large cohorts,** [Caitlin F. Harrigan](#), Ching Yeung Lam, Danian Chen, Christian Lai, Rod Bremner, Hartland W. Jackson, Kieran R. Campbell, *bioRxiv*
- 3. Measuring scientific capabilities of language models with a systems biology dry lab,** Haonan Duan\*, Stephen Zhewen Lu\*, [Caitlin F. Harrigan](#), Nishkrit Desai, Jiarui Lu, Michał Koziarski, Leonardo Cotta, Chris J. Maddison, *bioRxiv*, *Workshop paper at ICML 2025*, *Poster at NeurIPS 2025*
- 4. Bacterial ADP-heptose initiates a revival stem cell program in the intestinal epithelium,** Shawn Goyal, Cynthia X. Guo, Adrienne Ranger, Derek K. Tsang, Ojas Singh, [Caitlin F. Harrigan](#), Olga Zaslaver, Hannes L.

Rost, Herbert Y. Gaisano, Scott A. Yuzwa, Nan Gao, Jeffrey L. Wrana, Dana J. Philpott, Scott D. Gray-Owen and Stephen E. Girardin, *Cell Stem Cell*, 2025

**5. Whole slide Imaging Mass Cytometry allows the rapid profiling of the immune landscape of histopathologically aggressive prostate tumors**, Jennifer L. Gorman, Lydia Y. Liu, Jordan P. Hartig, Nikesh Parsotam, Amanda Khoo, Vladimir Ignatchenko, Sarah Asbury, Somi Afiuni, Ricardo Gonzalez, Michael J. Geuenich, [Caitlin F. Harrigan](#), Yuju Lee, Jianan Chen, Liang Lim, Qanber Raza, Peggi M. Angel, Kieran Campbell, Stanley K. Liu, Michelle R. Downes, Richard R. Drake, Thomas Kislinger, David King, Hartland W. Jackson, *AACR annual meeting proceedings*, 2024

**6. Regional mutational signature activities in cancer genomes**, Caitlin Timmons, Quaid Morris<sup>†</sup>, and [Caitlin F. Harrigan](#)<sup>†</sup>, *PLOS Computational Biology*, 2022

**7. Activating mTOR mutations are detrimental in nutrient-poor conditions**, Agata A. Bielska, [Caitlin F. Harrigan](#), Yeon Ju Kyung, Quaid Morris, Wilhelm Palm, and Craig B. Thompson, *Cancer Research*, 2022

**8. Considerations for visualizing uncertainty in clinical machine learning models**, [Caitlin F. Harrigan](#)<sup>\*</sup>, Gabriella Morgenshtern<sup>\*</sup>, Anna Goldenberg, and Fanny Chevalier, *Workshop paper at CHI 2021*

**9. Reconstructing evolutionary trajectories of mutation signature activities in cancer using TrackSig**, Yulia Rubanova, Ruian Shi, [Caitlin F. Harrigan](#), Roujia Li, Jeff Wintersinger, Nil Sahin, Amit Deshwar, and Quaid Morris, *Nature Communications*, 2020

**10. TrackSigFreq: subclonal reconstructions based on mutation signatures and allele frequencies**, [Caitlin F. Harrigan](#), Yulia Rubanova, Quaid Morris, and Alina Selega, *Pacific Symposium on Biocomputing*, 2020

#### ACADEMIC TALKS

**Mutational Signatures for DNA Damage and Misrepair**, BIRS: Mathematical Methods in Cancer Biology, Evolution and Therapy, *Invited talk* 05/23

**DAMUTA: Dirichlet allocation of mutations as a function of both damage and DNA repair**, Cold Spring Harbour Laboratory Meeting: Genome Informatics, *Selected Talk* 11/21

**TrackSigFreq: subclonal reconstructions based on mutation signatures and allele frequencies**, Pacific Symposium on Biocomputing, *Selected Talk, Poster* 01/20

#### ACADEMIC AWARDS

**Doctoral Student Fellowship Award** 09/22 - 09/25  
Data Science Institute, University of Toronto

**NSERC Postgraduate Scholarship - Doctoral** 09/22 - 09/25  
Natural Sciences and Engineering Research Council of Canada

**Graduate Research Award** 03/24 - 05/24  
Mitacs, in partnership with UKRI

**Queen Elizabeth II Graduate Scholarship in Science & Technology** 09/22  
Province of Ontario

**Ontario Graduate Scholarship** 09/21 - 09/22  
Province of Ontario

**ACM SIGHPC Computational & Data Science Fellowship** 07/20 - 07/22  
Special Interest Group on High Performance Computing of the Association for Computing Machinery

**JXTX foundation Genome Informatics Scholarship** 08/21

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|--------------------------------------------------------------------------------------------|-------|
| James P. Taylor Foundation for Open Science                                                |       |
| <b>General Motors Women in Science and Mathematics Award</b>                               | 09/20 |
| University of Toronto                                                                      |       |
| <b>NIH Conference Travel Fellowship</b>                                                    | 11/19 |
| International Society for Computational Biology                                            |       |
| <b>The Audrey Taylor Award for co-curricular contribution to the New College community</b> | 07/17 |
| University of Toronto                                                                      |       |

## GRANTS

|                                                                       |       |
|-----------------------------------------------------------------------|-------|
| <b>Emergent Data Sciences Program Grant</b>                           | 09/24 |
| Data Science Institute, University of Toronto                         |       |
| <i>Awarded for CrossTALK drug discovery bootcamp</i>                  |       |
| <b>Student initiative fund</b>                                        | 02/18 |
| University of Toronto                                                 |       |
| <i>Awarded for Bioinformatics and Computational Biology Hackathon</i> |       |
| <b>Event grant</b>                                                    | 02/18 |
| New College Student council                                           |       |
| <i>Awarded for Bioinformatics and Computational Biology Hackathon</i> |       |

## ACADEMIC SERVICE

|                                                                                                                                                        |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Journal peer review:</b> Genome Biology (2023), Genome Medicine (2021), Nature Cancer (2024), PLOS Computational Biology (2023 x2), iScience (2021) |  |
| <b>Conference referee:</b> CODEML @ ICML (2025)                                                                                                        |  |
| <b>Conference program committee:</b> MLCB (2019), Pacific Symposium on Biocomputing (2020)                                                             |  |

## PROFESSIONAL TRAINING

|                                               |       |
|-----------------------------------------------|-------|
| <b>The Carpentries certified instructor</b>   | 10/23 |
| <b>Oxford ML summer school (health track)</b> | 07/23 |

## STUDENT SUPERVISION

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| <b>Kiki Zhang</b> , Combined BS/MSE Student; Biomedical Engineering. Research internship via Computational Biology Student Program at MSKCC, <i>Topic: Mutational signatures in the context of branching evolution</i>                                                                                                    | 06/23 - 08/23 |
| <b>Fedir Zhydok</b> , BS Student; Computer Science. Artificial intelligence in medicine, global classroom program at University of Toronto, <i>Topic: Identifying metastatic tumours from mutational signatures</i>                                                                                                       | 09/22 - 12/22 |
| <b>Caitlin Timmons</b> , BA Student; Statistical and Data sciences, Biology. Research internship via Computational Biology Student Program at MSKCC. Went on to a Research Technician position at Dana-Farber Cancer Institute., <i>Topic: Modelling spatial distribution of mutational signatures in cancer genomes.</i> | 05/21 - 08/22 |
| <b>Haritha Lakshmanan</b> , Highschool Student. Independent study at MSKCC. Went on to a combined BA/MD at Brooklyn College., <i>Topic: Automatic discovery of mutations predictive of survival in breast cancer patients</i>                                                                                             | 05/20 - 11/20 |

## GUEST LECTURES

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Feature engineering &amp; data loaders</b> , CrossTALK drug discovery bootcamp                                                                                                                      | 06/25 |
| <b>Exploring &amp; Explaining Data in the Wild</b> , AI & Data Science Post-Graduate Program, Loyalist College, <i>Invited by Prof. Peter Papadakos as an annual guest lecturer in 2023 &amp; 2024</i> | 03/23 |

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|----------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Data Collection &amp; Analysis</b> , PRISM research & mentorship program, University of Toronto,<br><i>Invited by Prof. Sadia Sharmin</i> | 02/22 |
| <b>Environmental &amp; Life Sciences Workshop Series</b> , STEMHub Foundation                                                                | 01/20 |
| <b>R for bioinformatics workshop</b> , Global Society for Genetics and Genome Biology                                                        | 01/20 |

#### OTHER TALKS

|                                                                                                                                                                                        |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>Machine Learning for Cancer Genomics</b> , Artificial Intelligence in Healthcare Society Case<br>Competition, York University, <i>Invited keynote</i>                               | 03/24 |
| <b>Finding the ‘I’ in science</b> , ACM Canadian Celebration of Women in Computing, <i>Selected<br/>Workshop</i>                                                                       | 10/22 |
| <b>Undergraduate research opportunities: how to find them and make them work for you</b> , UofT<br>Bioinformatics and Computational Biology Student Union, <i>Invited by the BCBSU</i> | 02/20 |
| <b>How to hack your degree</b> , Computer Science Student Union, University of Toronto, <i>Invited by the<br/>CSSU</i>                                                                 | 05/19 |

#### PROGAM & EVENT ORGANIZATION

|                                                                                                               |               |
|---------------------------------------------------------------------------------------------------------------|---------------|
| <b>Program Organizer</b><br>UofT Graduate Application Assistance Program                                      | 09/21 - 01/25 |
| <b>Project Manager</b><br>STEMHub Foundation, Toronto, Canada                                                 | 10/20 - 05/21 |
| <b>Volunteer Coordinator</b><br>Computer science grad visit day, University of Toronto                        | 03/20         |
| <b>Founder and treasurer</b><br>Bioinformatics and Computational Biology Student Union, University of Toronto | 05/18 - 05/19 |
| <b>Program information session</b><br>UofT Bioinformatics and Computational Biology Student Union             | 06/17 - 06/19 |
| <b>Planning executive</b><br>Bioinformatics and Computational Biology Hackathon 2018                          | 03/18         |

#### TEACHING ASSISTANT POSITIONS

*Unless otherwise noted, school is University of Toronto*

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|----------------------------------------------------------------------------------|---------------|
| <b>CSC2541: Topics in Machine Learning: AI for Drug Discovery</b> , Head prep TA | 04/25 - 09/25 |
| <b>CrossTALK drug discovery bootcamp</b> , Head TA                               | 01/25 - 12/25 |
| <b>JSC370: Data Science II</b>                                                   | 01/23 - 05/23 |
| <b>JSC270: Data Science I</b>                                                    | 01/23 - 05/23 |
| <b>STA313: Data Visualization</b>                                                | 09/22 - 12/22 |
| <b>JSC370: Data Science II</b>                                                   | 01/22 - 05/22 |
| <b>CSC197: Privacy in the Age of Big Data Collection</b>                         | 09/21 - 12/21 |
| <b>STA4273: Minimizing Expectations</b>                                          | 01/21 - 05/21 |
| <b>CSC197: Privacy in the Age of Big Data Collection</b>                         | 09/20 - 12/20 |
| <b>JSC270: Data Science I</b>                                                    | 01/20 - 05/20 |
| <b>CSC373: Algorithm Design, Analysis &amp; Complexity</b>                       | 09/19 - 12/19 |

#### MENTORSHIP ACTIVITIES

As part of my ongoing commitment to supporting students at all levels and background in engaging with computational biology, I make an effort to be available to provide guidance and resources to students, with a particular

focus on creating an inclusive environment that fosters diverse perspectives and experiences. In addition to being a mentor through the organized programs listed here, I set aside ~2h/month for by-request 30 minute meetings.

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|------------------------------------------------------------------------------|---------------|
| <b>Graduate Application Assistance Program</b> , University of Toronto       | 09/21 - 01/25 |
| <b>Computer Science Alumni Mentorship Program</b> , University of Toronto    | 01/23 - 05/23 |
| <b>Statistical Science Alumni Mentorship Program</b> , University of Toronto | 10/21 - 05/22 |
| <b>ProjectX machine learning research competition</b> , UofT AI club         | 09/20 - 02/21 |
| <b>Her Code Camp</b> , Toronto, Canada                                       | 08/20 - 08/21 |
| <b>Department of Statistics Mentorship Program</b> , University of Toronto   | 10/18 - 05/19 |
| <b>SPROUT Peer Mentorship Program</b> , University of Toronto                | 09/16 - 05/19 |
| <b>New College E-Mentorship Program</b> , University of Toronto              | 09/16 - 05/19 |

#### **HACKATHONS**

|                                                                             |       |
|-----------------------------------------------------------------------------|-------|
| <b>Silver medal</b> , Toronto Bioinformatics Hackathon 2024                 | 09/24 |
| <b>Silver medal</b> , American Statistical Association DataFest Competition | 05/18 |

#### **COMMUNITY ACTIVITIES**

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|---------------------------------------------------------------------------------------------|-----------------|
| <b>Webmaster</b> , Project Galiwatch                                                        | 03/20 - present |
| <b>Graduate student ambassador</b> , Computer science grad visit day, University of Toronto | 03/25           |
| <b>Judge</b> , Lunenfeld-Tanenbaum Research Institute Summer Poster Session                 | 07/24           |
| <b>Webmaster</b> , IMERSS Xetthecum ecocultural digital mapping project                     | 10/23 - 12/24   |
| <b>Volunteer</b> , Meals on wheels                                                          | 06/23 - 09/23   |
| <b>Volunteer</b> , Billion oysters project                                                  | 05/21 - 09/21   |
| <b>Communications and Marketing Executive</b> , UofT Women in Computer Science              | 09/16 - 05/17   |

Last updated: Mar 2026